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Automated Multi-Channel Video Editing via Multimodal Analysis Using Deep Neural Networks

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I. INTRODUCTION

An Automated Editing System designed to produce a refined video by editing **multi-channel input** footage, such as conversations, discussions, and podcasts. Traditionally, human video editors have performed these complex tasks, by carefully combining multiple inputs into a single cohesive video. Developing an end-to-end AI solution capable of replicating this task requires well-defined data that encompass the nuanced behaviors inherent in the editing process. However, such data are exceedingly rare and difficult to obtain. Recent video editing systems have attempted to automate specific tasks within the editing process. For example, [**?**] performs video editing by using visual and textual metadata. However, this approach requires paired data where text metadata is provided for sub-clips under controlled conditions, making it unsuitable for general video editing scenarios. Additionally, [**?**] proposes a method for evaluating the quality of frames in an automated video editing system. Yet, visual analysis alone may be insufficient for audio-visual materials like videos, and the lack of subjective evaluation factors requires additional verification for practical applications. Similarly, [**?**] introduces a rule-based AI system developed to autonomously perform shot framing, sequencing, and participant selection by receiving wide-angle video streams as input for specific formats such as live panel shows, utilizing both visual and auditory analysis. While it proposes an efficient workflow improvement for live streams, it

does not incorporate the concept of processing multi-channel inputs, differing from our approach. Furthermore, the "Text-Based Editing" feature included in Adobe Premiere Pro [**?**] supports automatic rough cut generation along with text transcription by performing speech recognition and speaker diarization on the master audio within the video. However, in the context of multi-channel input video editing, automation for matching the transcriptions and segments to their corresponding input channels has not been achieved. These examples illustrate that existing video editing systems focus on automating only specific tasks within the editing process. They often lack a holistic approach that integrates both visual and auditory elements to create a seamless and professional final product from multi-channel inputs.

To address this challenge, we focus on the objective of generating a fully edited video using only the unedited footage from each channel by replicating and integrating the complex decision-making processes performed by human editors using advanced artificial neural networks. For practical and efficient editing, we incorporate sophisticated neural networks in both the visual and auditory domains. Our proposed automated multi-channel input video editing system addresses these limitations by integrating advanced neural networks in both the visual and auditory modalities, providing a more streamlined and fully automated editing experience. We perform inter-channel analysis at the frame level by integrating facial recognition neural network models and image classification models that classify the speaking state in the recognized facial

regions, and for the auditory approach, we integrate speech recognition and speaker diarization models to perform text transcription of audio, sentence-level segmentation, and speaker diarization. Based on this comprehensive analysis, we enable automatic editing of multi-channel video inputs by finely synchronizing visual and auditory elements.

The main contributions of our paper can be summarized as follows:

- Automated editing of multi-channel input video sources.
- A method that does not depend on well-curated datasets capturing the complex and subtle aspects of human editing behavior or well-defined data that encompass the nuanced behaviors inherent in the editing process.
- Integration of editing information from audio and visual modalities.

II. RELATED WORK

Recent advancements in automated multi-channel video editing have increasingly leveraged deep neural networks to enhance multimodal analysis and editing processes. Key research areas include **Score-Based Editing**, which utilizes quantitative metrics to select optimal shots for improved video coherence and quality. **Selecting Shots Using Virtual View**, where virtual camera techniques adjust shot levels and angles based on objects or performers of interest. And **Audio-Based Editing**, which analyzes and transcribes audio to detect events or speech segments that inform editing decisions. Additionally, significant progress in **Automatic Speech Recognition** (ASR), **Speaker Diarization**, and **Object Detection and Image Classification** has contributed to more accurate and efficient analysis, further enhancing automated editing capabilities through the use of deep neural networks.

Score-Based Editing focuses on enhancing the editing process by evaluating and selecting shots or sequences based on quantitative metrics or predefined quality scores. Several studies have contributed to this area, providing frameworks and methods to automate video editing through score-driven evaluation. For instance, [**?**] proposes a data-driven approach for automatic video editing, where editing decisions are guided by learned metrics derived from datasets. This study highlights the importance of understanding human aesthetic preferences and embedding them into machine learning models to

improve the quality and coherence of edited videos. Similarly, [**?**] introduces a framework for automated image and video quality assessment tailored for computational video editing. By utilizing perceptual quality metrics, this approach identifies high-quality shots and filters out undesirable footage, ensuring a more polished and professional outcome in the automated editing pipeline. Furthermore, [**?**] presents SB-VQA (Stack-Based Video Quality Assessment), a framework designed to assess video quality using a stack-based approach. The method provides accurate quality scores for video sequences, aiding in video enhancement tasks and ensuring that the selected shots meet predefined quality thresholds for effective editing. This approach demonstrates strong potential in optimizing editing workflows, particularly in scenarios where maintaining high video quality is critical.

Selecting Shots Using Virtual Shot is an automated editing method that adjusts or changes the level and angle of shots based on performers or objects of interest within a single wide-angle recorded video. Studies such as [**?**] and [**?**] propose systems that automatically generate cinematic edited videos from footage captured by a single static wide-angle camera, utilizing virtual camera techniques and gaze data. Both studies are structured into two main stages: virtual camera generation and shot selection. Additionally, [**?**] performs shot sequencing using face detection and tracking, facial landmarking, visual speaker detection, and audio event detection. The system evaluates shot levels and selection based on the number of detected individuals and an

analysis of the volume of speech from speaking individuals.

Audio-Based Editing involves analyzing the audio contained in a video, transcribing it, and detecting events or speech segments for use in video editing. [**?**] proposed a system that automatically generates audiovisual metadata by utilizing speech recognition, event identification on audio, and image recognition in medical education content, and automatically edits multiple videos into one cohesive video based on this metadata. Adobe Premiere Pro's "Text-Based Editing" feature [**?**] is an editing tool that automatically converts video audio into text, and when the user edits the text, the changes are immediately reflected on the timeline. This allows users to quickly search for specific words or phrases and easily remove unnecessary parts, greatly improving the efficiency of video editing.

Automatic Speech Recognition(ASR) involves converting speech signals into text by analyzing patterns within the speech signal to recognize linguistic units such as phonemes, words, and sentences. This process typically includes phoneme modeling, where the speech signal is used to model phonemes, and language modeling, which combines recognized phonemes into coherent textual output to ensure accurate transcription. DeepSpeech [**?**] is a prominent ASR model utilizing recurrent neural networks (RNNs) for end-to-end speech recognition. A key innovation of DeepSpeech is the elimination of manual feature extraction, relying instead on deep learning to directly convert speech to text. By employing the Connectionist Temporal Classification (CTC) loss

function [**?**], DeepSpeech aligns the speech signal with the target transcription, effectively handling speech sequences of varying lengths. Wav2Vec 2.0 [**?**] introduces a novel approach by learning representations directly from raw audio waveforms without requiring labeled data for pre-training. The model employs self-supervised learning to capture general speech patterns from unlabeled data, enabling fine-tuning on smaller amounts of labeled data for specific tasks. Utilizing transformers to model long-range dependencies in speech, Wav2Vec 2.0 demonstrates high effectiveness in low-resource settings and environments with scarce labeled data. Whisper [**?**] focuses on robust speech recognition across various languages, accents, and noisy environments. It employs a Transformer-based architecture adept at capturing long-range dependencies in speech. Designed to handle a wide range of real-world scenarios, including multilingual speech and noisy backgrounds, Whisper is highly versatile for diverse speech recognition applications. Each of these models in DeepSpeech, Wav2Vec 2.0, and Whisper offers unique strengths. DeepSpeech's end-to-end RNN approach simplifies the ASR process. Wav2Vec 2.0's self-supervised learning from raw audio enhances performance in low-resource environments. And Whisper focus on robustness makes it suitable for real-world, multilingual applications.

Speaker Diarization involves partitioning an audio stream into segments according to speaker identity, effectively addressing the question "who spoke when?". It is widely used in multi-speaker environments such as meetings,

podcasts, and interviews. Traditional methods rely on feature extraction techniques like Mel-Frequency Cepstral Coefficients (MFCCs), followed by segmentation and clustering algorithms such as K-Means or Gaussian Mixture Models (GMMs) to group similar speech segments. Recent approaches have adopted deep learning-based techniques, improving performance by leveraging speaker embeddings and advanced clustering methods. For instance, x-vectors extracted from deep neural networks provide more robust representations of speakers in diarization tasks. End-to-end approaches, such as End-to-End Neural Speaker Diarization (EEND), have further streamlined the diarization process by jointly learning speaker embeddings and performing speaker segmentation within a single model. This approach significantly enhances accuracy and effectively handles challenging scenarios, including overlapping speech. A notable diarization toolkit is *Pyannote.audio* [**?**] [**?**] features for handling complex multi-speaker scenarios.

Object Detection and Image Classification are pivotal areas of computer vision, which have been greatly advanced by deep learning models. Convolutional neural networks (CNNs) have revolutionized these tasks with effective feature extraction and classification techniques. Models such as AlexNet [**?**], VGGNet [**?**], and ResNet [**?**] have established hierarchical feature extraction methods, making CNNs the standard for image-based tasks. In Object detection, CNN-based approaches such as Faster R-CNN [**?**], YOLO [**?**], and SSD [**?**] have excelled in real-time feature extraction and bounding box esti-

mation. Vision transformers (ViTs) [**?**] introduced a paradigm shift in image classification by using self-attention mechanisms to model global dependencies. By considering an image as a series of patches, ViTs capture long-range relationships across the entire image, performing well on tasks that require global context. However, it is computationally intensive. *Swin transformer* [**?**] enhances ViTs by implementing a hierarchical architecture that moves windows for efficient self-attention. Because this design balances local and global information processing, the Swin transformer is effective for dense prediction tasks such as face detection and segmentation, resulting in improved accuracy and efficiency. Overall, the evolution of these networks has led to significant progress in face detection and image classification: CNNs excel at local feature extraction, while ViTs and Swin transformer provide superior global modeling.

III. PROPOSED METHOD

For multi-speaker, multi-channel content, an intuitive approach was necessary to make automated editing feasible. This is primarily because datasets containing multi-speaker, multi-channel videos paired with their fully edited versions are extremely rare, as they would need to represent highly specific editing scenarios. The lack of such datasets makes it challenging to apply standard supervised learning techniques effectively for automated video edit-

ing.

To overcome this limitation, we focused on replicating the editing actions of human editors through artificial neural networks for each modality. Rather than relying on curated datasets of edited videos, we aimed to substitute the decision-making processes of human editors with deep learning models capable of processing audio, visual, and other relevant data independently.

In our approach, ASR was employed to transcribe spoken content from audio channels, and Speaker Diarization was utilized to identify who was speaking at each point in time. Speaking State Estimation helped determine when individuals were actively contributing to the discussion. Visual features, including face detection and facial expression analysis, were also leveraged to provide contextually appropriate edits. Each neural network model was trained to handle a specific aspect of the editing process, allowing us to create a system capable of understanding the dynamics of multi-speaker interactions. By addressing each modality audio, visual, and derived outcome independently and then integrating these insights, our proposed system can adapt to diverse editing scenarios without relying on pre-labeled datasets. This approach, integrating multiple neural networks, ultimately enabled our system to replicate the nuanced and context-aware decisions typical of human editors, thereby reducing manual intervention while preserving the quality and coherence of the edited video content.

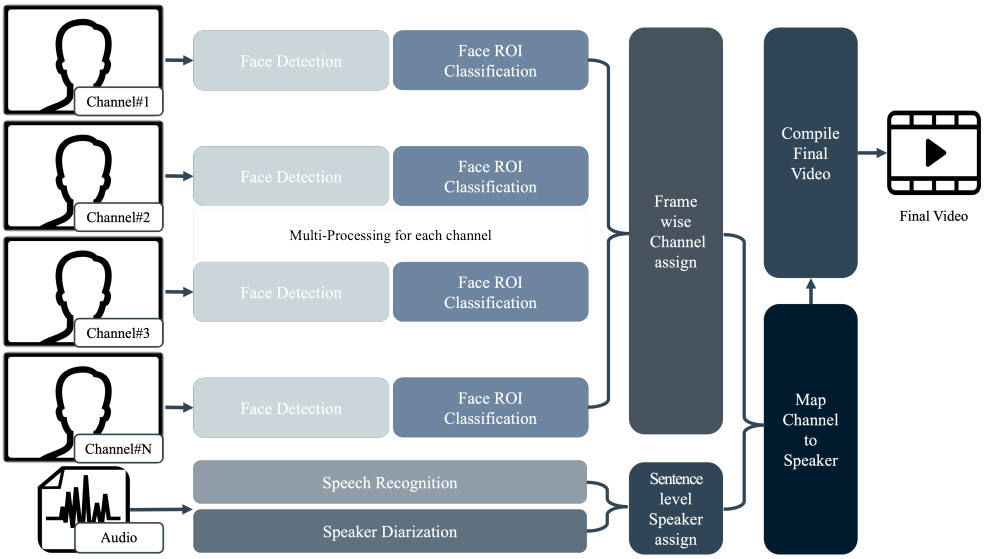

Figure 3.1: Overall System Architecture

3.1 Audio Modality

To address the challenges of diverse speakers, languages, and accents, we adopted a weakly supervised speech recognition model. As outlined in [**?**], weakly supervised models offer strong general performance without requiring extensive fine-tuning, making them more adaptable than unsupervised approaches. This allows our system to generalize better across varying speaking styles and environments, which is crucial for multi-speaker and multi-language content.

Our method leverages weak supervision to handle unstructured audio data, reducing the need for human intervention during training. This makes the system scalable and resilient to noisy conditions, allowing it to perform well

across different domains. The approach enhances downstream tasks such as Voice Activity Detection, ASR, and Speaker Diarization contributing to a robust automated video editing pipeline.

To improve accuracy, we integrated *WhisperX* [**?**], which builds upon the Whisper. WhisperX provides both word-level and sentence-level time alignment, crucial for synchronizing video edits with spoken content. However, due to padding issues in the underlying Whisper model, the timestamps generated for sentences are not always precise. This padding introduces slight offsets, particularly for sentence boundaries, which can lead to synchronization issues in video editing.

To address this limitation, we incorporated Pyannote.audio [**?**,**?**] for speaker diarization, allowing us to refine the sentence-level timestamps by accurately identifying the active speaker for each segment. By applying Pyannote.audio' s speaker diarization in conjunction with WhisperX outputs, our system achieves more reliable alignment, especially in multi-speaker scenarios.

The audio processing begins with WhisperX transcription, which outputs approximate sentence-level timestamps. For a sentence S occurring over a time interval T, WhisperX generates the pair (S, T), where:

$$T = (t_{\text{start}}, t_{\text{end}})$$

This timestamp provides a coarse alignment of spoken content, but due to

the padding issue, additional refinement is necessary. In this refined alignment stage, we use Pyannote.audio to perform speaker diarization.

The task of assigning speaker labels to each transcribed word involves combining diarization segments, which include speaker identity and temporal boundaries $(t^D_{start}, t^D_{end}, speaker)$, with transcription segments or words $(t^S_{start}, t^S_{end}, w)$. The goal is to determine the speaker who most likely uttered each word. This is achieved by evaluating the temporal overlap between transcription intervals and diarization intervals. For each transcription segment or word S and each diarization segment D, we compute the temporal intersection as follows:

$$\text{intersection} = \min(t^D_{end}, t^S_{end}) - \max(t^D_{start}, t^S_{start})$$

and the temporal union as:

$$\text{union} = \max(t^D_{end}, t^S_{end}) - \min(t^D_{start}, t^S_{start})$$

If the intersection is positive, it indicates that the diarization segment D and the transcription segment S overlap in time, suggesting that the speaker associated with D may be responsible for at least part of the utterance represented by S. By aggregating these intersection values for all candidate speaker segments, the speaker with the highest total intersection is assigned to the tran-

scription segment or word. This process can be first applied at the segment level and subsequently at the word level within each segment, ensuring a fine-grained alignment of sentences and speaker identities. Thus, through these calculations and assignments, each transcribed sentence is ultimately annotated with the most likely speaker, enabling more nuanced analyses of conversational structure and speaker-dependent content.

3.2 Visual Modality

The visual approach involves estimating the speaking state based on the facial region of the speaker. This method requires accurate detection of the face region, which serves as the basis for analyzing facial expressions and movements related to speaking. To achieve this, we perform face region detection for each frame, allowing us to track and analyze even subtle facial dynamics in real time. By applying fine-grained face detection and speaking state estimation on a frame-by-frame basis, we can mitigate errors that might arise due to sudden or unexpected changes in the video, such as occlusions or shifts in facial position.

To address scenarios where the speaker's face is partially or fully obstructed or where sudden video changes occur, we incorporate an error-handling mechanism into our system. Specifically, when the face detection model fails to locate a face or when occlusion is detected, the system interpolates the

bounding box B using temporal smoothing methods based on previously detected frames. This ensures continuity and minimizes disruptions caused by missing detections. Additionally, a confidence score threshold is applied to the output of the detection model, ensuring only reliable face detections are utilized. In cases where the confidence score falls below the threshold, the system flags the frame for further analysis or fallback handling.

We employ MediaPipe' s *Face Detection* [**?**] [**?**], which provides robust and efficient face region detection. Specifically, we use Face Detection, a lightweight and high-speed model that identifies the bounding box of the face within each frame. This approach is well-suited for real-time applications due to optimization, allowing us to detect faces accurately even under challenging conditions such as varying lighting or partial occlusions. The Face Detection outputs a bounding box for each frame, represented as:

$$B = (x_{\text{min}}, y_{\text{min}}, x_{\text{max}}, y_{\text{max}})$$

where B is the bounding box surrounding the detected face in each frame, with $(x_{\text{min}}, y_{\text{min}})$ and $(x_{\text{max}}, y_{\text{max}})$ representing the coordinates of the top-left and bottom-right corners of the box, respectively. If the bounding box B cannot be reliably detected due to occlusion or other issues, the system estimates B using interpolation and context from adjacent frames. This enables us to extract the face region of interest (ROI), which is crucial for further analysis,

even in the presence of challenging conditions. Using this detected face ROI, we proceed to estimate the speaking state of the speaker. For this purpose, we utilize a Swin transformer [**?**] configured for binary classification. The extracted face ROI is fed into the Swin transformer, which predicts whether the speaker is in an active speaking state or not. The classification process can be defined as:

$$S = f(B)$$

where S represents the speaking state (active or inactive), and f is the function implemented by the Swin transformer, taking the bounding box B as input. This approach allows our system to operate effectively in real-time, providing reliable results even in complex multi-speaker and dynamic video scenarios. The model uses a final classification head to produce a label indicating whether the speaker is actively speaking. Specifically, the classification head includes fully connected layers to output a probability value between 0 and 1, which determines if the speaking state is active or inactive:

$$P_{\text{speaking}} = \sigma(W_{\text{fc}} \cdot F + b_{\text{fc}})$$

where W_{fc} and b_{fc} are the weights and bias of the fully connected layer, F represents the features extracted by the Swin transformer, and σ is the softmax

function. The final decision rule is:

$$\text{Speaking State} = \begin{cases} \text{Active} & \text{if } P_{\text{speaking}} > 0.5 \\ \text{Inactive} & \text{otherwise} \end{cases}$$

This architecture benefits from Swin transformer' s ability to efficiently handle visual features and capture both local and global context within the face ROI, thus significantly improving the accuracy of speaking state detection. The lightweight nature of Face Detection and the efficiency of the Swin transformer allows our system to operate at a high frame rate, ensuring that the speaking state is estimated accurately even in real-time scenarios. By capturing details effectively and consistently throughout the video, Face Detection and Swin transformer contribute to the robustness and responsiveness of our speaking state estimation approach. By leveraging Face Detection to extract the face ROI, followed by Swin transformer for classification, we achieve a robust and effective solution for estimating the speaking state. The error-handling mechanisms integrated into the pipeline, such as temporal smoothing, confidence thresholding, and fallback strategies, reduce errors caused by occlusion or sudden video changes. Additionally, the system incorporates multiprocessing to process video streams frame by frame, enabling efficient handling of multi-channel video inputs in parallel. For each frame, the system analyzes multiple channels and selects the channel with the highest proba-

bility of containing the active speaker. This frame-level multiprocessing enhances accuracy and responsiveness.

3.3 Integration of data from audio and visual modalities

In multi-channel audio-visual systems, accurate speaker-to-channel mapping is crucial for content processing, especially in environments with multiple speakers. Manual mapping is labor-intensive and prone to errors, resulting in inefficiencies. To address these challenges, we propose an automated algorithm leveraging transcription and multi-channel probability data to enhance the accuracy of speaker-to-channel mapping.

Our approach begins by utilizing transcription data to determine the time intervals when each speaker is active. These intervals, denoted as t_{start} and t_{end}, are converted into corresponding frame indices based on the video frame rate (fps). The start and end frames, f_{start} and f_{end}, are computed as follows:

$$f_{\text{start}} = \lfloor t_{\text{start}} \times \text{fps} \rfloor, \quad f_{\text{end}} = \lfloor t_{\text{end}} \times \text{fps} \rfloor$$

These frame indices help synchronize transcription data with video frames, ensuring accurate alignment between spoken sentences and their visual representation. For each frame within the interval $[f_{\text{start}}, f_{\text{end}}]$, the algorithm assigns the channel with the highest probability. Denoting the probability of a channel c being correct at frame f as $P(c, f)$, the selected channel c_{max} for each frame

is given by:

$$c_{\text{max}}(f) = \underset{c}{\operatorname{argmax}}\, P(c, f)$$

After processing all frames, the most frequently assigned channel is chosen for each speaker, ensuring consistency throughout the segment. This can be expressed as:

$$c_{\text{final}} = \underset{c}{\operatorname{argmax}}\, \text{Count}(c_{\text{max}}(f))$$

where $\text{Count}(c_{\text{max}}(f))$ represents the frequency of each channel within the segment.

To further enhance consistency, a smoothing step is applied to handle short, inconsistent speaker segments caused by noise or brief interruptions. Let the length of a segment be L, and the predefined threshold be T. The speaker-to-channel mapping process can be broken down into three essential steps:

1. **Time-to-Frame Conversion**: Transcription time intervals are converted into frame indices to facilitate precise alignment between audio and visual components.

2. **Probability-Based Mapping**: Each frame within a speaking segment is analyzed, and the channel with the highest probability is selected. The final channel for the segment is determined by taking the mode (most frequently occurring channel) of the selected channels, ensuring

stability and reliability.

3. **Smoothing**: Post-processing is applied to short segments to maintain consistency and avoid jerky transitions.

Algorithm 1 Algorithm for Mapping Speaker to Maximum Probability Channel

```
Input: multi_channel_data, transcription_data, fps (default: 30)
Output: speaker_max_prob_mapping
Initialize speaker_channel_map as a dictionary with list values
  Initialize last_known_channel as None

foreach segment in transcription_data do
    if segment is a dictionary then
        speaker ← segment["speaker"]
          if speaker is None and last_known_channel exists then
            speaker ← last_known_channel
        end
        else if speaker is not None then
            last_known_channel ← speaker
        end
        start_frame ← int(segment["start"] × fps)
          end_frame ← int(segment["end"] × fps)

        foreach frame from start_frame to end_frame do
            if frame exists in multi_channel_data then
                max_prob_channel ← multi_channel_data[frame]
                  Append max_prob_channel to
                  speaker_channel_map[speaker]
            end
        end
    end
end
foreach speaker, channels in speaker_channel_map do
    if channels is not empty then
        most_common_channel ← the most frequent channel in channels
          speaker_max_prob_mapping[speaker] ← most_common_channel
    end
    else
        speaker_max_prob_mapping[speaker] ← "No channel data"
    end
end
return speaker_max_prob_mapping
```

Algorithm 1 is designed to automate the mapping process of assigning speakers to the most probable speaking channel. It efficiently iterates through each segment of the transcription data, extracting and analyzing the information to determine the corresponding channel. The algorithm identifies the most probable channel for each speaker while systematically managing cases where the speaker information might be missing. It does this by tracking the last known speaker and seamlessly handling continuity in speaker assignments, thus ensuring consistent and reliable channel mapping.

By minimizing potential disruptions caused by missing or ambiguous data, Algorithm 1 provides a robust approach to speaker-to-channel mapping. It ensures that variations in speaking activity are accounted for and maintains the overall accuracy of the mapping process. This approach is particularly beneficial in dynamic environments where speaker transitions and gaps in data are frequent, as it emphasizes maintaining the integrity and reliability of the mapping results.

Algorithm 2 Algorithm for Smoothing Short Segments in Speaker Data

```
Input: speaker_data (list of speakers), threshold (default: 15)
Output: smoothed_speaker_data (list of speakers)
Convert speaker_data to an array  n ← length of speaker_data
segment_start ← 0 // Variable to store the start index
    of the current segment
for i ← 1 to n do
    if i = n or speaker_data[i] ≠ speaker_data[segment_start] then
        segment_length ← i - segment_start
        if segment_length ≤ threshold then
            left_speaker ← speaker_data[segment_start - 1] if segment_start >
             0 else None  right_speaker ← speaker_data[i] if i < n else None
            if left_speaker ≠ None and left_speaker = right_speaker then
                Set speaker_data[segment_start : i] ← left_speaker
            end
            else if left_speaker ≠ None then
                Set speaker_data[segment_start : i] ← left_speaker
            end
            else if right_speaker ≠ None then
                Set speaker_data[segment_start : i] ← right_speaker
            end
        end
        segment_start ← i // Update the start index for the
            next segment
    end
end
return speaker_data as a list
```

To complement this, Algorithm 2 focuses on refining the data by smoothing out short, noisy segments. By leveraging neighboring speaker data, the algorithm avoids abrupt transitions and creates a more seamless flow. This approach is particularly effective in noisy, multi-speaker environments where interruptions are common, ensuring high-quality data processing.

Overall, the integration of these algorithms automates the speaker-to-channel mapping process, significantly enhancing efficiency and accuracy. The approach is scalable and operates in real-time, even with large video files, mak-

ing it well-suited for automated video editing workflows. By handling missing data, applying smoothing techniques, and employing probabilistic mapping, the system provides a robust solution for complex multi-speaker scenarios.

IV. EXPERIMENT

4.1 Experimental Setup

4.1.1 Audio Modality

For processing the audio modality, we employ a pre-trained automatic speech recognition model, whisper-large-v2, along with a pre-trained speaker diarization framework, Pyannote.audio. These components are integrated using WhisperX, enabling robust extraction of linguistic features, precise temporal alignment at the word and sentence levels, and reliable speaker identification.

4.1.2 Visual Modality

For processing the visual modality, we employ Face Detection to extract Face ROIs. To estimate speaking states, we train the Swin transformer model using the dataset described below.

- Dataset: We utilized the Lip Reading dataset [**?**], which comprises speech videos recorded with various speakers and filming angles. We

selected frames corresponding to utterance times specified in the labels, as well as those outside these intervals, and applied Face Detection to identify and extract Face ROIs. Each Face ROI was assigned a binary label indicating whether it represented an active utterance or silence. This process yielded a large-scale dataset to facilitate robust model training and evaluation. The training set contained approximately 7,975,028 utter face images and 6,740,797 silence face images, while the validation set consisted of about 872,377 utter face images and 526,647 silence face images.

- Training Details: We trained and validated the Swin transformer on the previously described dataset for 10 epochs, using a learning rate of 5e-4 and the Adam optimizer, in conjunction with a Noam learning rate schedule to ensure stable convergence. The training process was distributed across six A6000 GPUs with PyTorch' s DataParallel framework. As a result, the model achieved a validation accuracy of 95.88%.

4.2 Evaluation

To evaluate our system, we utilized two multi-channel video samples, which are samples used in actual broadcasts in South Korea. The first sample consists of interview content featuring two speakers, while the second

contains conversation content featuring four speakers. We primarily focus on the four-speaker conversation sample, for which we have labels representing channel transitions throughout the entire video. Additionally, this sample includes segments corresponding to sub-intervals of the full duration, which represent human editing units.

We compare the labeled data with the output generated by our system at the frame level. By quantifying this match rate under various criteria and providing visualizations, we assess the overall effectiveness of our system.

Table 4.1: Match Rates Across Multichannel Data

Data	Lip	Face	Lip+Speech	Face+Speech
2-channel video	0.6429	0.7918	0.8460	**0.8460**
4-channel video	0.6618	0.7353	0.8972	**0.8977**

Table 4.2: Effects of Frame Smoothing

Data	Threshold 15	Threshold 20	Threshold 30	Threshold 40
2-channel video	0.8382	0.8440	0.8460	**0.8469**
4-channel video	0.8763	0.8865	**0.8977**	0.8951

Table 4.3: Match Rates Across Different Segment Lengths

Interval (sec)	Total Matching Frames	Total Frames	Match Rate
$0\text{ s} \leq \text{Length} \leq 5\text{ s}$	3,405	4,335	0.7857
$5\text{ s} < \text{Length} \leq 10\text{ s}$	5,352	5,940	0.9000
$10\text{ s} < \text{Length} \leq 20\text{ s}$	2,928	4,425	0.6617
$20\text{ s} < \text{Length} \leq 30\text{ s}$	3,600	3,720	0.9677
$30\text{ s} < \text{Length} \leq 40\text{ s}$	7,904	8,580	0.9211
$40\text{ s} < \text{Length} \leq 50\text{ s}$	2,524	2,550	0.9906
$50\text{ s} < \text{Length} \leq 60\text{ s}$	1,648	1,650	0.9988
$60\text{ s} < \text{Length} \leq 70\text{ s}$	2,010	2,010	**1.0000**
$70\text{ s} < \text{Length} \leq 80\text{ s}$	2,163	2,220	0.9743
$80\text{ s} < \text{Length} \leq 90\text{ s}$	2,670	2,670	**1.0000**
Total	34,204	38,100	**0.8977**

Table 4.4: Match Rate for each specific span

Span	Duration	Match Rate
Span 0	(0,200)	0.89
Span 1	(200,400)	0.85
Span 2	(400,600)	0.93

Span	Duration	Match Rate
Span 3	(600,800)	0.87
Span 4	(800,1000)	**0.96**
Span 5	(1000,1200)	0.86
Span 6	(1200,1270)	0.81
Overall	(0,1270)	**0.8977**

Table 4.5: Match Rate for each channel

Channel	Match Rate	Match Frame	Frame Count
#0	0.94	12,244	13,065
#1	**0.99**	14,190	14,340
#2	0.87	3,322	3,840
#3	0.81	2,528	3,105
#W	0.51	1,920	3,750
Overall	**0.8977**	34,204	38,100

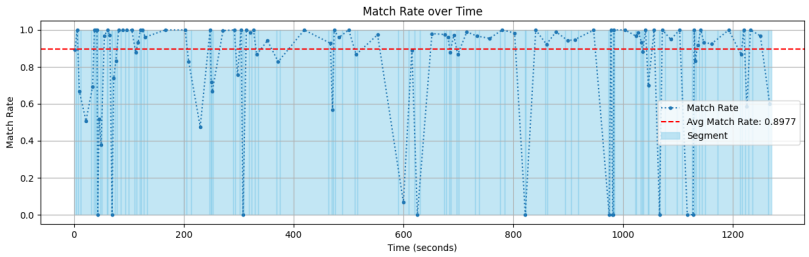

Figure 4.1: Match Rates Over Time

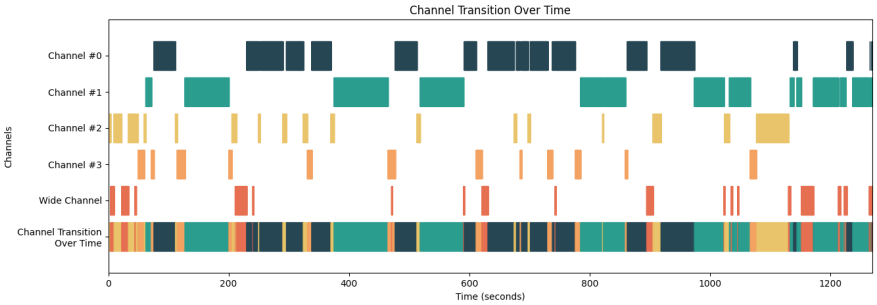

Figure 4.2: Channel Transition Over Time

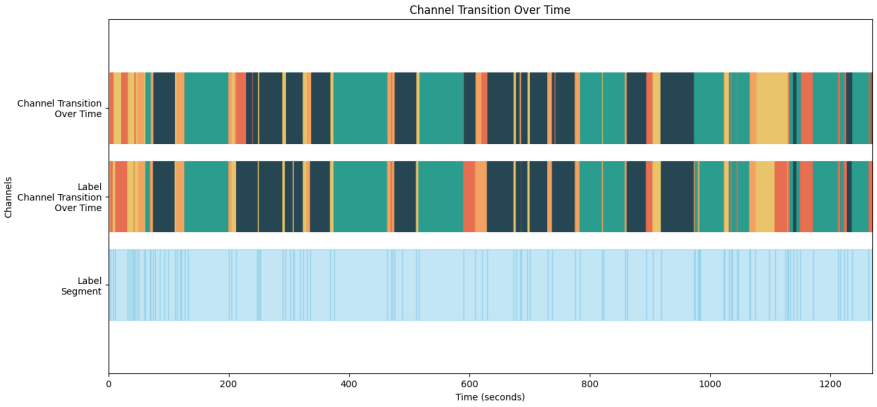

Figure 4.3: Channel Transition & Labels, Segments Over Time

V. RESULTS

Table 4.1 presents the match rates for two types of multichannel data based on different analytical approaches. Specifically, lip analysis and face analysis were conducted by detecting regions of interest (ROIs) across all frames and classifying the speaking state within each detected ROI. When lip analysis was performed independently, the match rates were 0.6429 and 0.6618 for the two types of multichannel data, respectively. In contrast, when face analysis was performed independently, the match rates were 0.7918 and 0.7353, respectively. Furthermore, integrating speech analysis with both visual analyses resulted in significantly higher match rates, achieving 0.8460 and 0.8977 for the two datasets. This substantial improvement highlights the effectiveness of combining visual and audio modalities in enhancing the accuracy of speaking state classification.

Table 4.2 shows the match rates obtained by varying the threshold for frame smoothing while adopting the method that integrates face analysis and speech analysis. When the threshold was set to 30 frames, equivalent to 1 second or more, high match rates of 0.8469 and 0.8977 were observed for the respective sample. This indicates that applying a frame smoothing threshold of 1 second or longer is highly effective in enhancing match rates. By allowing sufficient time for transitions, the system maintains contextual consistency and reduces the likelihood of mismatches during abrupt channel changes.

Tables 4.3, 4.4, and 4.5 evaluate the performance on 4-channel video sample. Specifically, Table 4.3 analyzes the match rates by categorizing subsegments based on their lengths. Short segments exhibited a minimum match rate of 0.6617, whereas long segments achieved a perfect match rate of 1.0. This demonstrates the system' s superior performance in longer segments, where it can maintain higher accuracy. Additionally, Table 4.4 presents match rates by setting the span length to 200 seconds for video progression. The highest match rate of 0.96 was observed in the mid-segments (800 seconds and 1000 seconds), while the final segments (1200 seconds and 1270 seconds) had a match rate of 0.81. Table 4.5 shows the match rates based on speaker differentiation. Channel #1, with the highest number of speaking frames (14,340 frames or 478 seconds), exhibited the highest match rate of 0.99. In contrast, the Wide channel #W, which was applied in situations where decision-making was not possible without analysis, showed a lower match rate of 0.51. This lower rate is likely due to the absence of analysis for the Wide channel.

5.1 Quantitative Analysis

- **Match Rates Across Multichannel Samples:** As shown in Table 4.1, lip-based analysis alone achieves lower match rates compared to face-based analysis. This suggests that facial features are more reliable indicators of speaking states than lip features alone. Moreover, the integra-

tion of audio features with visual analyses leads to substantial improvements, demonstrating that combining multiple modalities enhances the robustness of speaking state classification.

- **Effects of Frame Smoothing:** The findings in Table 4.2 highlight the effectiveness of applying a frame smoothing threshold of at least 1 second. Such smoothing mitigates abrupt channel transitions, maintaining contextual consistency and thereby improving overall match rates.

- **Influence of Segment Length:** Analysis of varying segment lengths in Table 4.3 shows that longer segments consistently achieve higher match rates. This pattern suggests that extended temporal contexts allow the system to more accurately capture and maintain the speaking state, leading to increased reliability.

- **Performance on Fixed-Length Segments:** Table 4.4 demonstrates that segmenting the video into fixed intervals (e.g., 200-second segments) results in generally high match rates. Such an approach provides a stable temporal framework, further enhancing the system' s accuracy and robustness.

- **Channel-Level Performance:** Table 4.5 indicates that channels with a higher frequency of speech activity attain higher match rates. In contrast, the Wide channel, employed as a fallback in cases lacking suf-

ficient analytical cues, shows a lower performance. This underscores the importance of available analysis opportunities and clear signals for reliable state classification.

5.2 Qualitative Analysis

- **Match Rates in Sub-Segments:** Fig. 4.1 illustrates the match rates within sub-segments across all intervals. Each point on the graph represents the frame match rate at the midpoint of the blue windows (sub-segments). This visualization reveals fluctuations in match rates over time, offering insights into how the system performs in different portions of the sample. By examining these points, one can assess the consistency and reliability of the system' s performance under various contexts.

- **Visualization of Channel Transitions:** Figs. 4.2 and 4.3 provide a comprehensive visualization of channel transitions throughout the entire duration. The system' s outputs align closely with the labeled data, exhibiting a high degree of consistency. Notably, high match rates appear during speaking intervals and longer-duration segments, indicating effective channel selection during critical periods.

  Furthermore, the system demonstrates adaptability even in intervals with overlapping active channels which is an inherently complex sce-

nario. While it captures the general trend of human decision-making in regions with frequent transitions, the system struggles to fully replicate the intricate considerations that human editors apply, especially when overlapping speech and rapid changes occur in short segments. This underscores the need for further improvements in handling highly complex editing decisions.

VI. DISCUSSION

In this study, we performed both audio and visual analyses to derive automatic editing decisions and applied these findings to multi-channel video inputs. Our results show that integrating these two modalities enables more effective channel selection. We also observed that editing is more effective when the speaking state extends over longer intervals, indicating that maintaining segment coherence benefits the overall editing outcome. Furthermore, adopting a sentence-level editing approach yielded natural-looking results.

For future work, analyzing the transcribed text of each utterance may offer a more sophisticated basis for editing. We believe our system effectively determines each participant' s active state and automatically edits multi-channel content in a highly efficient manner, laying the groundwork for more advanced, context-aware editing frameworks.

VII. CONCLUSION

In this study, we developed an efficient system that replicates and automates the intricate editing process carried out by human editors on multi-channel video inputs, leveraging advanced neural networks. Specifically, our approach integrates solutions for automatic speech recognition, speaker diarization, face detection, and face ROI classification. We then propose an algorithm that combines this information to produce automated editing decisions.

To validate our system, we compared its outputs against human-edited videos across multiple sample sets, conducting analyses from various perspectives. Our findings confirm the effectiveness of integrating multiple modalities. In particular, we observed a 0.8977 match rate for the four-channel video, demonstrating the viability of our approach for this specialized task. We believe our method offers a novel and robust framework for automatic editing of multi-channel video inputs, laying the groundwork for further enhancements in future research.

Automated Multi-Channel Video Editing via Multimodal Analysis Using Deep Neural Networks

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(*Abstract*)

Editing multi-channel videos in the broadcast domain, such as conversations, discussions, and podcasts, traditionally relies on complicated human-intensive processes. AI-based automatic editing presents a significant challenge due to the limited availability of well-curated datasets that represent the complex and subtle aspects of human editing behavior. This paper introduces an automated multi-channel video editing system that generates a fully edited video using only the original video footage from each channel. Our approach can replace and integrate the complex decision-making actions of human editors by employing advanced artificial neural networks in both the visual and

auditory domains. In the visual domain, we conduct frame-level inter-channel analysis by integrating a face recognition neural network and an image classification neural network that assesses the speaking states within recognized facial regions. In the auditory domain, we incorporate speech recognition and speaker identification models to perform text transcription, sentence-level segmentation, and speaker identification. By finely synchronizing and integrating comprehensive data from both domains, our system enables automatic editing of multi-channel video inputs, resulting in a coherent and professional final product. Experimental results demonstrate that our system achieves a 0.89 frame match rate with manually edited videos for a video source with four channels. Our method simplifies the editing process and establishes a foundation for future advancements in automated media production, enabling AI-driven tools to adapt seamlessly to the evolving needs of content creators.

심층 신경망을 사용한 멀티모달 분석을 통한 자동화된 멀티채널 비디오 편집

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(초록)

대화, 토론, 팟캐스트 등 방송 분야에서 멀티채널 영상을 편집하는 작업은 전통적으로 복잡하고 인력이 많이 투입되는 과정을 거쳐 왔습니다. 인간 편집자의 복합적이고 미묘한 편집 방식을 충분히 반영할 수 있는 잘 정제된 데이터셋이 제한적이기 때문에, 인공지능(AI) 기반 자동 편집은 여전히 큰 도전 과제로 남아 있습니다.

본 논문에서는 각 채널에서 얻은 원본 영상만을 활용하여 완전하게 편집된 결과물을 생성하는 자동 멀티채널 영상 편집 시스템을 제안합니다. 이 방법은 시각 및 청각 영역 모두에서 고도화된 인공신경망을 사용함으로써, 인간 편집자의 복잡한 의사 결정 과정을 대체하고 통합할 수 있습니다.

시각 영역에서는 얼굴 인식 신경망과 이미지 분류 신경망을 결합하여, 인식된 얼굴 영역에서 발화 상태를 평가하고 프레임 단위로 채널 간 분석을 수행합니다. 청각 영역에서는 음성 인식과 화자 식별 모델을 통해 텍스트 전사, 문장 단위 분할, 화자 식별을 수행합니다.

두 영역에서 추출된 종합적 데이터를 정교하게 동기화하여 통합함으로써, 멀티채널 영상 입력을 자동으로 편집하고, 일관성이 높고 전문적인 최종 결과물을 만들어냅니다. 실험 결과, 4개 채널을 가진 영상 소스에 대해 수동 편집본과 비교했을 때 0.89의 프레임 일치율을 달성했습니다.

본 연구는 편집 과정을 단순화하고 향후 자동화된 미디어 제작의 발전을 위한 토대를 마련함으로써, AI 기반 도구들이 변화하는 콘텐츠 제작자의 요구에 유연하게 대응할 수 있도록 지원합니다.